

A-level

ENVIRONMENTAL SCIENCE

Questions by topic: Ozone depletion - Set 1

Mark scheme

AQA spec 3.2.1.3; The physical environment; textbook pp. 108-113

Marking guidance

- Marking points are separated by bullets; each bullet is worth 1 mark unless stated.
- Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	• UV light splits diatomic oxygen (O₂) into two monatomic oxygen atoms (O) • Monatomic oxygen reacts with diatomic oxygen to form ozone (O₃) • UV light also splits ozone into O₂ and O; the reactions form a dynamic equilibrium which absorbs UV (converting it to chemical energy / heat)	3
01.2	• Useful: boiling points near ambient temperature so easily liquefied under low pressure (aerosol propellants, fridge / air-conditioning coolants); dissolve grease but do not damage electrical components (solvents); non-flammable; non-toxic • Persistence: chemically stable / unreactive in the troposphere so they remain long enough to be carried up to the stratosphere • Dissociation by UV: in the stratosphere higher UV levels break the carbon-chlorine bonds releasing chlorine free radicals • Chlorine reacts with monatomic oxygen (and ozone), preventing ozone formation; chlorine is regenerated so one atom destroys many ozone molecules; bromine / iodine act similarly	6
01.3	• UVB energy is absorbed by cells and breaks up biological molecules / DNA causing mutations, skin cancer, cataracts, sunburn (human health) • Damage to leaf tissue and reduced photosynthesis / crop yields (plants) • Damage to marine algae, phytoplankton, corals and planktonic larvae (marine organisms), reducing productivity and food webs	3
01.4	• An area where the total ozone column falls below 220 Dobson Units (normal is about 300 DU) - a level never recorded before CFC pollution	1
01.5	• Concentration varies with altitude and the stratosphere is not normally sampled; Dobson Units measure the total column thickness instead	1
01.6	• The British Antarctic Survey at Halley detected higher UV at ground level, showing ozone above had decreased (satellites above the stratosphere cannot measure downward UV directly)	1

Question	Marking guidance	Marks
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 108-109; AQA spec 3.2.1.3.	
02.1	- Minimum ozone fell steeply from about 225 DU in 1979 to about 95 DU in the mid 1990s - Since then it has fluctuated around 100-115 DU with a slight recovery / no further decline	2
02.2	$(225 - 95) / 225 \times 100$ = 57.8% (accept 58%)	2
02.3	- Stratospheric temperatures over Antarctica are the lowest on Earth so stratospheric clouds / ice crystals form - Ice crystal surfaces allow reactions that convert the chlorine released from CFCs into chlorine molecules (Cl₂) during the dark winter - The polar vortex (rotating winds) prevents mixing with the rest of the atmosphere, keeping temperatures low and chlorine concentrated - When sunlight returns in spring (September) it splits the chlorine molecules into free radicals that destroy ozone; levels recover by December as the vortex breaks down	4
02.4	- Ground and satellite UV / ozone measurements show depletion but not the chemistry / correlation is not causation - Air samples collected from the stratosphere by helium balloons and high-flying research aircraft detected chlorine and chlorine monoxide, confirming the Rowland-Molina hypothesis	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 110-111; AQA spec 3.2.1.3.	
03.1	- Manufacture and use of CFCs and other ozone-depleting substances (ODSs) phased out then banned - HCFCs to be phased out (by 2030); some essential uses (eg halon fire extinguishers in aircraft) still permitted - A fund to help (developing) countries implement the Protocol; legally binding targets (unlike the 1985 Vienna Convention)	3
03.2	- Aerosols: pump-action sprays / roll-on deodorants (alternative process), or hydrocarbon propellants (propane, butane) - flammable / fire risk - Refrigerants: HCFCs (less stable so break down in the troposphere, but still some ozone depletion) then HFCs (no chlorine) - more expensive, less effective, powerful greenhouse gases	4
03.3	- CFCs are drained from the appliance and incinerated / broken down at high temperature - Products are CO₂ and acidic gases (HCl, HF) which are neutralised with crushed lime	2
03.4	CFCs are persistent, so those already released continue to reach the stratosphere and destroy ozone for many years; some ODSs (eg dichloromethane) are not banned	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 112-113; AQA spec 3.2.1.3.	

Question	Marking guidance	Marks
04.1	Levels of response Level 3 (7-9 marks): Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. Level 2 (4-6 marks): Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. Level 1 (1-3 marks): Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. 0 marks: No relevant content. Indicative content (credit other relevant, accurate points) • Stages of scientific method illustrated by ozone: identification (Antarctic UV data), hypothesis (Rowland-Molina 1974), data collection / analysis (Dobson units, satellites, balloons confirming CIO), proposed solutions, enactment (Vienna 1985, Montreal 1987) • Evidence of success: ozone hole area and depth stabilised and beginning to shrink; CFC concentrations in the atmosphere falling; UV levels not rising further; global ozone loss limited to about 4% • Reasons for success: international recognition of serious consequences (skin cancer); agreement of nearly every country; legally binding targets with a fund for developing countries; relatively few producers / products; alternatives (HCFCs, HFCs, pump sprays, hydrocarbons) already available and cheap; no lifestyle change needed • Limits: persistence of CFCs means recovery takes decades; some ODSs still used; HFC replacements are greenhouse gases • Climate change contrast: greenhouse gases come from almost every human activity (energy, transport, agriculture, industry) in every country; fossil fuels underpin economies so control requires major lifestyle and economic change • Agreements (Kyoto, Paris) have weaker / non-binding targets, disputes between developed and developing nations; emissions still rising; natural fluctuations, time delays and tipping points make cause-effect less clear to the public • Judgement: ozone control was a technical substitution problem; climate change is a systemic one, so the ozone success is only partly transferable	9
04.2	• Vienna Convention for the Protection of the Ozone Layer - no legally binding goals	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 108-113, 105-106; AQA spec 3.2.1.3, 3.2.1.2.	
05.1	• C	1
05.2	• Stratospheric ozone is natural and beneficial: absorbs UV, protecting living organisms • Tropospheric ozone is a secondary pollutant (from NO_x photochemistry) that is toxic to plants and humans and is a greenhouse gas	2
05.3	• HCFCs are less chemically stable / contain hydrogen so they are broken down in the troposphere • So most do not persist long enough to reach the stratosphere where UV would release their chlorine	2

Question	Marking guidance	Marks
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 108-112; AQA spec 3.2.1.3.	